Fine-tuning of universal machine-learning interatomic potentials for 2D high-entropy alloys

C. Zhou[1] H. P. Komsa[1]

[1]Microelectronics Research Unit, Faculty of Information Technology and Electrical Engineering, University of Oulu, P.O. Box 4500, Oulu, FIN-90014, Finland

High-entropy alloys (HEAs) and their two-dimensional counterparts have attracted increasing interest due to their tunable properties and catalytic potential, while their chemical complexity poses significant challenges for first-principles modeling. In this work, we discuss our recent progress on combining density functional theory (DFT) and universal machine-learning interatomic potentials (uMLIPs) to study two-dimensional transition metal sulfide HEAs. We assess the applicability of several leading uMLIPs and show that direct use of universal models is insufficient for reliably describing key energetic quantities such as mixing energies. We then present effective fine-tuning strategies based on systematically generated random and enumerated structures, which substantially improve model robustness and accuracy. We validate the fine-tuning strategies across three distinct structural databases constructed from different alloy configurations, showing that enumeration-based training sets generally lead to more efficient and stable models compared to random structure selection. Using the experimentally synthesized (Mo,Ta,Nb,W,V)S_2 system as a case study, we integrate the fine-tuned models with Monte Carlo simulations to analyze phase decomposition behavior and observe phase separation behavior that follows the temperature-dependent Gibbs free energy trends of the quinary alloy and its possible decomposition products. Overall, the study demonstrates the potential of fine-tuned uMLIPs for modeling chemically complex alloys, while emphasizing the importance of carefully designed training strategies for achieving reliable predictions.